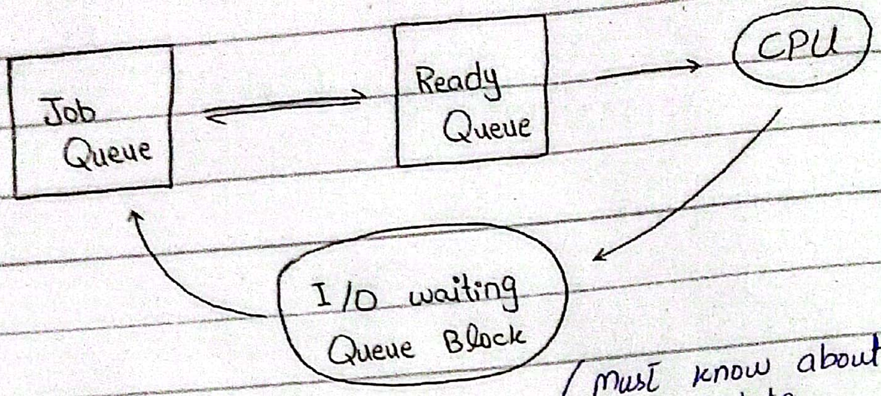
Operating System

Chapter #01

- Definition
- Types
    - Batch OS
    - Time sharing OS
    - Real Time OS
    - Embedded OS
    - Android / Mobile OS
    - General purpose OS
        - ↳ e.g., Windows, Mac, Linux
- Interrupts
- Kernel

Chapter # 02

- Process
- Process Attributes
- Process States
- States Transition
- Process Control Block
- Process Schedulling
    - long term
    - short term
    - Medium term



(Must know about every state)

- Process creation
  - Theory & fork();
- Process Termination
  - Theory & exit() or abort();
- Context switching
- Independent & Cooperative Process
- Inter-process Communication (IPC)
  - Shared memory
  - Message Passing
    - ↳ Direct vs Indirect

Chapter # 04

- Objectives of Scheduling
  - Maximize CPU utilization
  - " Resource Usage
  - " Throughout
  - Minimize waiting time
  - " Respone time

- Minimize turn-around time
- Fairness
- Age-ing
- Uniformity
- CPU I/O Burst time
  - ↳ CPU bound
  - ↳ I/O bound
- Timer
- Dispatcher

- Times
  - Arrival
  - Response
  - Waiting
  - Burst
  - Turnaround
  - Completion
- Algorithm types:-
  - Preemptive
  - Non-preemptive
- FCFS (Non-preemptive)
- SJF (Non-preemptive)
- SJF (Preemptive) aka (SRTF)
  (Shortest Remaining Time First)
- Priority Scheduling (Non-preemptive)
- " " (Preemptive)
- Round Robin

- Multi-level Queue
- Multi-Level Queue vs. Multi-level feedback Queue

Chapter #03

- Threads
- Difference b/w Process & Thread